

# Making Lava Lamps



## Equipment:

- Flask or empty water bottle.
- (Vegetable, cooking, baby) oil.
- Water.
- Food colouring.
- Fizzing tablets.

## Method:

1. Fill the flask (just greater than half-way) with oil.
2. Fill the remainder with water.
  - a. The oil floats and the water sinks to the bottom of the jar under the oil - why do you think this is?
3. Add a few drops of food coloring.
  - a. The coloring is water based and so will sink with the water.
4. Then break a fizzing tablet into a few small pieces and drop them into the flask **one at a time.**

# Concept 1: Density (this slide is not very dense)

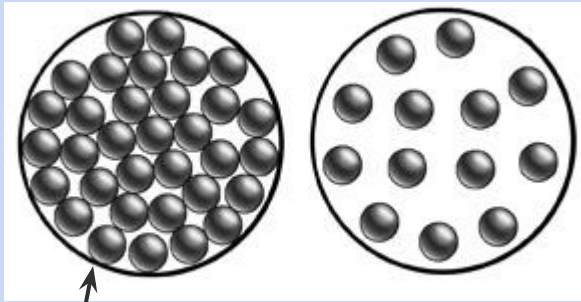
- Density is a measure of how compact a substance is.
- It is defined as “mass per unit volume.”

$$\rho = \frac{m}{V}$$

$\rho$  = density

$m$  = mass

$V$  = volume



Same volume more  
particles therefore  
**more dense.**

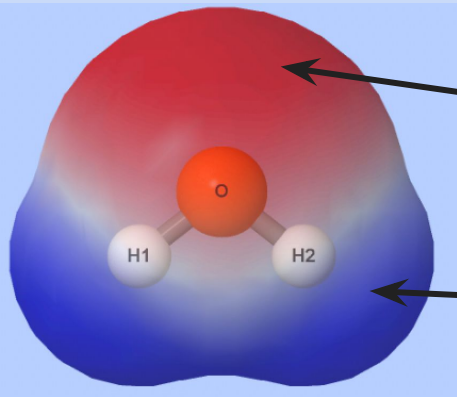
- Oil is **less dense** than water.
  - Therefore if you measured an equal volume of water and oil, the oil would have a lower mass.
- This is why water sinks to the bottom and oil floats to the top.

**FUN FACT!** Density is affected by temperature. The hotter the liquid, the less dense it will be. This is because the molecules have more energy and spread apart.

# Concept 2: Polarity

Water is a polar molecule, but what does this mean?

- Electrons orbit in clouds around atoms.
  - Electrons have a **negative** charge and protons have a **positive** charge therefore, they are **attracted** to one another.
- When hydrogen bonds with oxygen to form  $\text{H}_2\text{O}$  (water) hydrogen's electrons are drawn towards oxygen.
- This is because oxygen has **more protons** than hydrogen.
- This leads to a lop-sided electrical charge that attracts other atoms.



Electrons drawn to this end.  
This end now has a small **negative** charge.

This end now has a small **positive** charge due to the lack of electrons.

- Oil molecules are **not polar**.
- They are completely neutral.
- Therefore they are **not** attracted to the water molecules.
- This is why water and oil **don't** mix and why the oil bubbles move through the water.